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Duration, trading volume and the price impact of trades in an emerging futures market

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This draft: October 2007

Abstract

This paper examines the price impact of trading intensity on an emerging futures market. Utilizing a novel volume-augmented duration model of price discovery, the intensity effect is decomposed into liquidity and information components for the MexDer 28-day interest rate futures contract. We find that the duration between transactions exerts a positive influence on price changes. Increases in order flow and trade volume exert a positive and negative influence on price changes, respectively. The liquidity and information coefficients exhibit a contrasting effect on returns. The dominance of the liquidity component suggests that it is the liquidity traders who dictate the time to trade.

JEL Classification: G15, G14, C32
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1 Introduction

Theoretical models generally explain trading activity in financial markets as arising from either information or liquidity considerations. Understanding market microstructure and the relationship between prices, trading volume and duration (the time between consecutive trades), is critical to determining both the information and liquidity aspects of trading activity and the underlying value of a financial asset. This paper examines the price impact of trading intensity utilizing a novel volume-augmented duration model of price discovery. Empirical analysis is conducted on more than six years of high-frequency (tick by tick) data taken from an emerging futures market, the Mexican derivatives market (MexDer), a pure limit order book market.

The analytical focus is upon the relationship between key trading process variables, specifically; order flow, duration, and trading volume and subsequent asset price changes. Prior research customarily focuses upon the price impact of one or two specific trade process variables in isolation, rather than their combined effects. For example, Madhavan et al. (1997) concentrate solely on the order flow variable assuming trade size to be constant. They argue both that this allows for a more parsimonious model estimation and also, if markets are truly anonymous, informed traders will split up their large trades, implying that trade direction should be more informative than trade size. In contrast, for De Jong et al. (1996) trade size constitutes one of the key components of the bid ask spread, while several theoretical market microstructure models (Easley and O'Hara (1987), Engle (2000) and Manganelli (2005)) maintain that trading volume reflects market relevant information.

The contribution of this paper is twofold. First, we explicitly include trading volume as a component of the trade direction indicator. This choice can be rationalized along several dimensions. Easley and O'Hara (1987) argue that considerations relating to both liquidity and optimal trading strategy dictate that informed traders may sometimes choose to engage in high volume intensive trading, while others segment their trades in order to generate a larger number of informationally based trades over a period of time. Thus, trade volume conveys information relating to market conditions and may have a direct effect upon prices. Duration has also variously been considered as an indicator of trading intensity and liquidity, estimated with various trading-activity related measures. We believe that volume is also potentially relevant as one such activity measure. Indeed, Engle (2000) and Manganelli (2005) model volume as a

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stochastic process when estimating the trading impact of duration. Moreover, De Jong et al. (1996) provide evidence that transaction size (and by association trading volume) is positively related to measured pricing effects. On the basis of these arguments, we extend the Ben Sita and Westerholm (2006) model of the price effects of trading intensity by incorporating the all important volume variable. The extended model is then estimated in the context of an emerging futures market.

This constitutes our second contribution. We are among the first to examine the price effects of duration for an important developing (emerging) financial market. Prior research in this area almost universally focuses upon highly liquid equity, derivative or foreign exchange spot markets of developed economies, where traditionally trades tend to be more frequent and the level of liquidity greater than that in developing financial markets.[1,2] We also maintain that the analysis of a futures market offers further potential insights on the price impact of the components of the bid-ask spread, and how both public and private information contributes to the determination of market equilibrium prices than can be obtained solely from spot market studies. This is because many spot markets are characterized by institutional rigidities such as position limits on traders or short-sales constraints which by their very nature mediate the response of asset prices to informational shocks. The present analysis seek to understand if certain theoretical predictions of existing market microstructure models continue to hold in a futures market setting, where traders can take long or short positions with equal facility. Moreover, trading volume in a futures market is likely to be an important factor impacting price variability, since the time to contract maturity is known to influence futures trading patterns as, for example, traders seek to unwind or rollover positions close to contract maturity. We examine trading on the MexDer using the 28-day equilibrium interbank interest rate futures contract (Tasa de Interes Interbancaria de Equilibrio or TIIE28), the most actively traded futures contract listed on the MexDer. In October 2006, the TIIE28 accounted for 93% of MexDer trading volume and 98.9% of the open interest on the exchange.[3] Indeed, volume increased 188.5% on the previous year and the TIIE28 ranks as the 4th most actively traded derivatives contract world-wide.

[1] See, *interalia*, Dufour and Engle (2000), Holder et al. (2004), Chung et al. (2005) and Ben Sita and Westerholm (2006).
[2] Zhong et al. (2004) investigate the role of price discovery between the Mexican futures and spot market using daily data on the IPC equity contract.
[3] MexDer.

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Overall, the MexDer ranks as the 8th largest derivative exchange by volume globally.[4] We provide a key insight into the microstructure of this significant market and important contract.

We demonstrate that both duration and trading volume play a significant role in the pricing behavior of Mexder TIIE28 futures contracts. First, our investigations involving transaction duration shows that the transient and permanent components of duration exert positive and negative influences on price changes, respectively. Overall, the net effect is similar to the evidence provided by Ben Sita and Westerholm (2006). In contrast to the results of Dufour and Engle (2000), we find that long duration has a greater impact on price changes than short duration. Second, our analysis of trading volume suggests that price changes tend to decrease when there is an increase in trade volume. Additionally, increases in order flow exert a positive influence on price changes, albeit to a lesser extent than the influence of duration.

The present paper is structured as follows. In the next section we present an overview of the existing literature. Section three discusses the econometric model and the methodology employed. Section four presents the empirical analysis and section five briefly concludes.

2 Literature Review

In theoretical microstructure models of financial markets, trading occurs either for information (information-based models) or for liquidity-motivated (inventory-based models) reasons. Information-based models focus on the adverse selection or asymmetric information costs arising from the private information of informed traders, and encompass both sequential trade models (for example Glosten and Milgrom, 1985 and Easley and O'Hara, 1987, 1992) and strategic models (for example, Kyle, 1985 and Admati and Pfleiderer, 1988). Sequential trade models examine the determinants of the bid-ask spread in a competitive environment with information distributed heterogeneously across agents. They are characterized by a probability structure in which prices act as informational signals and there is a learning problem confronting market participants. Strategic trade models maintain that private information generates incentives for agents to act strategically to maximize trading profits by selecting the timing and size of their trades. This strategic behavior might be affected by the trading mechanism (Kyle, 1985). These types of models generally induce patterns in trading activity, returns, and volatility.

[4] Holz (2007).

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Several comments are relevant in the context of this study. First, time *per se* is not explicitly accounted for in either class of models. Second, closer inspection reveals that these models differ in their treatment of informed traders. For example, Glosten and Milgrom (1985) maintain that informed traders will trade intensively as if immediacy provides their sole opportunity to trade, while in Kyle (1985) they choose to trade gradually so as not to reveal their private information to others. Third, it is clear that there is implicitly a time dimension present in these models because prices converge at different rates for uninformed and informed traders. Indeed, when comparing the above two models Back and Baruck (2004) demonstrate that informed (uninformed) traders trade with a lower (higher) trading rate (that corresponds to one time unit (the square root of that time unit)). Despite these differences, they find evidence that the Kyle (1985) and Glosten and Milgrom (1985) models actually capture the same underlying phenomenon, namely that trades move prices reflecting the possibility that the trader is better informed than the market at large. Easley and O'Hara (1992) further demonstrate the significance of time in price discovery in a framework where information flow is discontinuous as informed traders abstain from trading on occasions, and market makers equate the absence of informed traders to a lack of news. However, this finding holds only under the assumption that there is sufficient liquidity to allow informed traders to trade at their convenience, and that they do not exhibit patience in the sense of Kyle (1985).

In inventory models, the trading process is effectively a matching problem in which the dealer or market maker uses prices to balance their inventory. The key factors are the inventory position and the uncertainty about order flow. Dealers achieve inventory control by altering their bid and ask quotes to elicit the desired imbalance of buy and sell orders (see for example, Ho and Stoll, 1983 and O'Hara and Oldfield, 1986) thereby mediating deviations in orderflow. These deviations depend on the behavior of the market in the short run, so a dealer's effect on prices is temporary because their trades are independent of information, depending on trading costs, the dealer's previous position and net demand to the dealer.

Time is first explicitly accorded prominence and incorporated into market microstructure models in Diamond and Verrecchia (1987) and Easley and O'Hara (1992). In the sequential model of Diamond and Verrecchia (1987) set in the context of an equities market the commencement of trading coincides with one of two possible informational events, the arrival of good or bad news, observed by the informed traders. Upon receiving good news, an informed

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trader will always purchase the stock. If news is bad, a trader will wish to sell or alternatively sell short if they do not own the stock. If institutional rigidities dictate a trader informed of bad news faces a short-sales constraint there may be no trading. The implication is that informed traders will always trade unless they do not own the stock and/or short-sales constraints exist, the presence of which mitigates the adjustment of prices to private information, especially with respect to bad news. An uninformed trader is assumed to trade for reasons unrelated to the arrival of information. Accordingly, long durations are more likely to be associated with bad news.

Easley and O'Hara (1992) provide a different explanation for the role of time, relating trade timing to the existence of new information. Once again, an uninformed trader's decision to trade is independent of the existence of new information. However, an informed trader will only choose to transact if there is new information on the market, so a lack of trading can provide a signal of no price-relevant information arriving on the market. It follows that variations in trading intensity in Easley and O'Hara (1992) are closely associated with the change in the participation rate of informed traders. More generally, the receipt of new information may cause informed traders to trade more frequently, and hence their presence may quickly be ascertained by observing large volumes of trade (Easley and O'Hara, 1987). In the knowledge that this may move prices and quotes against themselves, restricting profit opportunities and correspondingly their incentive to trade, informed traders may choose to segment large volume trades, generating a larger number of smaller volume informationally-based trades. It follows that both trading intensity (which results in both short and long durations between trades), and trading volumes may provide information concerning the behaviour of market participants, in particular informed traders (Dufour and Engle, 2000). In our view, this justifies the explicit modeling of the impact of volume, which we undertake in this paper. Moreover, as a futures market operates without any short-sales constraints, futures prices are able to reflect good and bad news symmetrically. Thus, there is no institutionally-based rationale for the view that long duration is customarily associated with bad news. A pertinent question therefore is, in such a market is there evidence that information-based trades continue to exist at longer durations?

Empirical analysis of the operation of markets customarily employs time series data of a given frequency, where the variables of interest are separated by equally spaced fixed points in time. Alternatively stated, time is non-stochastic. With the availability of high frequency data, empirical work on microstructure issues can now investigate the importance of time by modeling

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it as a stochastic process. The necessity to consider the time dimension of the price process is stressed in much of the market microstructure literature (Easley and O'Hara, 1992; O'Hara, 1995; Easley et al., 1997). However, an issue arises when dealing with ultra-high frequency data as observations on duration, the time between consecutive transactions, are irregularly spaced. Engle and Russell (1998) suggest tackling this feature by explicitly modeling the duration between trades using an autoregressive conditional duration (ACD) model, allowing the time elapsed between two market events to convey meaningful market relevant information.

Duration, has been utilized to proxy several variables when testing market microstructure hypotheses. The two main theoretical predictions subject to empirical corroboration are those of Easley and O'Hara (1992), that short duration reflects information-based trading, and Diamond and Verrecchia (1987), that long durations lead to declining prices. Engle (2000) uses duration as a proxy for trading intensity to test these hypotheses obtaining results consistent with both predictions. Dufour and Engle (2000) formulate a VAR specification to analyze the price impact of trades adopting duration as a measure of liquidity, an important component of the bid-ask spread.[5] In this context, duration can be considered as a natural measure of the speed by which market liquidity is enhanced and prices incorporate new information. The results of the model reveal that short durations are associated with larger spreads, larger volumes, and a higher price impact of trades, which is consistent with Easley and O'Hara (1992) predictions. Finally, duration is also utilized as a measure of trading risk (Gouriéoux et al., 1999; Renault and Werker, 2002; Ghysels et al., 2004). In these studies, duration is modeled as a process capturing the risk associated with trading under both price and time uncertainty. Of particular relevance to this paper is Renault and Werker (2002), who decompose duration into a deterministic component, with a transient impact on returns which they relate to expected duration, and a stochastic component with permanent effects on return variance, capturing innovations in trading activity.

Examining the price effects of trade intensity, Madhavan et al. (1997) provide evidence showing that the trade indicator captures short-term deviations of the observed price from the fundamental (efficient) price. On the basis of a structural model, they show that the transitory price effects account for approximately 60.0 per cent of the total price variance. They

[5] Other studies that utilise duration as a proxy for liquidity are, for example, Engle and Lange (2001) and Engle and Lunde (2003).

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hypothesize that a number of variables can be considered to capture these effects. De Jong et al. (1996) provide evidence showing that the price effect increases with the transaction size, and is estimated to lie between 25.0 per cent and 60.0 per cent of the bid-ask spread. The relation between duration and returns is further investigated in Dufour and Engle (2000) who extend the Hasbrouck (1991) vector autoregression (VAR) model. Dufour and Engle (2000) provide evidence showing that prices, bid-ask spreads and price volatility all increase when traders observe short durations. This finding is consistent with the Easley and O'Hara (1992) prediction that market makers revise their prices upward in the presence of informed traders.

3 Econometric Methodology

3.1 Foundation for the model

There are two generic sources of information on which traders rely when conveying their beliefs about an asset's future value; public information obtained through news on fundamentals and private information that aggregates the information of informed traders. In contrast to public information, private information is first revealed through the order flow of informed traders, who buy (sell) when prices are below (above) fundamental value. Traders also learn from past price movements (Glosten and Milgrom, 1985), price duration (Easley and O'Hara, 1992) and price size (Easley and O'Hara, 1987). These three dimensions are important when characterizing the pricing process in financial markets. Earlier studies investigate these price dimensions using either the direction of trade (Glosten and Harris, 1988) or both the size and the direction of trade (Hasbrouck, 1991). Key to our analysis Dufour and Engle (2000) provide evidence supporting the hypothesis that informed trading is associated with short trade durations consistent with Easley and O'Hara (1992) who predict that market makers revise their prices upward in the presence of informed traders. Contrary to this proposition is the view that short durations are associated with the presence of liquidity traders who prefer to trade when the market is "thick" (Admati and Pfleiderer, 1988). When liquidity is high because of volatility, informed traders are not likely to trade at opening or at closings as expected. Therefore, when examining the role of time in the price discovery process, we utilize normalized duration measures with expectations about unity, to adjust for any intra-day seasonal patterns. We extend the Dufour and Engle (2000) empirical model by separating the component of duration associated with liquidity from

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that associated with information. Further, motivated by Renault and Werker (2002) who show that duration can be split into transient and permanent effects, we model the returns as a function of the order flow, the expected duration, duration innovations and the trade volume.

3.2 The Structural Model

In this section, we specify a simple model that decomposes the bid-ask spread into order-handling, liquidity and adverse selection costs utilizing the trade direction indicator, duration and a trade volume variable. The order handling cost is a fee charged by the liquidity supplier for standing ready to match buy and sell orders; inventory holding costs are associated with the risk of holding inventory stock, while adverse selections costs arise as a result of trading with informed agents. Several studies evaluate the relative importance of each component of the bid-ask spread by means of a cross-sectional decomposition of the spread into its respective components, operating on both cross-sectional stock portfolios and individual stocks. Glosten and Harris use the trade indicator variable to decompose the spread into permanent and transitory components. Stoll (1989) estimates the three components using a covariance based model for a cross section of NASDAQ stocks. Similarly, Roll (1984) infers the components of the bid-ask spread from the serial covariance properties of observed trade prices. Madhavan et al. (1997) suggest a quote-based model using order flow to decompose the spread into order handling and adverse selection costs, while Huang and Stoll (1997) use both the trading indicator and the trading volume to decompose the bid ask spread into all three components. The present model both modifies and extends the approach of Ben Sita and Westerholm (2006)[6] with respect to our inclusion of a trade volume variable, mediated by a trade direction indicator. The rationale for including trading volume is that previously outlined. Easley and O'Hara (1987) maintain trade volume conveys information relating to market conditions and may have a direct effect upon prices. Duration has variously been utilised as an indicator of trading intensity and liquidity, estimated with various trading-activity related measures. We maintain that volume is also potentially relevant in this respect. Finally, De Jong et al. (1996) provide evidence that transaction volume may be related to pricing impact. The initial model is:

[6] In turn they extend the Madhavan et al. (1997) model by including trade duration, decomposed into both permanent and transitory components

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$$m_t = m_{t-1} + \left[\phi_1 + (\phi_2 + \rho_q \theta_1) T_t + \phi_3 v_t)\right] q_t + e_t \quad (1)$$

where m_t is the unobservable (efficient) price at time *t*, such that $m_t = E\left[p_t^* \middle| \Phi_t\right]$; here p_t^* is the (log) fundamental value and m_t is the expected value of p_t^*. T_t is the transaction time, v_t the trade volume, q_t is the buy-sell trade direction indicator variable (taking -1 for seller-initiated trades, +1 for buyer-initiated trades, and 0 for trades occurring within the bid-ask spread), and e_t is a serially uncorrelated public information shock. Equation (1) decomposes the impact on the efficient price, m_t into five trading-related parameters: ϕ_1 captures the cost of processing a trade, ϕ_2 the speed at which transactions occur, θ_1 reflects private information revealed by order flow and trading intensity,[7] ϕ_3 reflects private information revealed by trading volume and ρ_q is the temporal dependence in the order flow resulting either from informed traders in their efforts to take advantage of incipient liquidity or noise traders unable to reverse the direction of trade.

While the efficient price, m_t is unobservable, the trade price, p_t, is observed albeit measured with error. Following decomposition of the duration variable into temporal and permanent effects (Renault and Werker, 2002), we obtain the first difference of equation (1), and substitute the trade price, p_t, as a proxy for the unobservable efficient price, m_t:

$$r_t = \phi_1 \Delta q_t + \phi_2 (T_t q_t - T_{t-1} q_{t-1}) + \rho_q \theta_1 (T_t q_t - T_{t-1} q_{t-1}) + \phi_3 (v_t q_t - v_{t-1} q_{t-1}) + \varepsilon_t + n_t - n_{t-1}$$

where $e_t - e_{t-1}$ is translated into $n_t - n_{t-1}$ which are serially correlated rounding errors (again proxying trade price for the efficient price, where n_t summarizes the effects of noise trading). Gathering like terms gives:

$$r_t = \left[\phi_1 \Delta q_t + (\phi_2 \psi_t + \rho_q \theta_1 C_t + \phi_3 z_t) q_t\right] + (\varepsilon_t + (n_t - n_{t-1})) \quad (2)$$

Where r_t is the continuously compounded log return in percent, ψ_t is the conditional duration process, C_t is the innovation in trading intensity which is defined as the covariance of the

[7] This variable is intended to capture the theoretical predictions of Easley and O'Hara (1992).

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standardized residual of duration, $(x_t/\psi_t)((x_t/\psi_t)-1)$. Δq_t is the residual in the order flow, z_t is the conditional change in volume. ε_t is the serially uncorrelated public information shock which is equivalent to the change in the unobservable fundamental value (price), $m_t - m_{t-1}$. $n_t - n_{t-1}$ is the serially correlated rounding error term which arises since the observable trade price, p_t, measures the unobservable efficient price, m_t, with error. Equation (2) implies that the return, r_t, reflects both the public information shock (through ε_t) and trading friction effects (through n_t) which are associated with the trade price, p_t.

Equation (2) decomposes actual duration, x_t, into a temporal component (measured by the expected duration, ψ_t) which captures liquidity effects and a permanent component (measured by the duration innovation component, C_t) which captures information effects. These two components are obtained from an ACD(1,1) model. Analogously the expected change in volume z_t is obtained from an ACV(1,1) model. In addition, the residual term is decomposed into a fundamental component (ε_t) and a transitory component $(n_t - n_{t-1})$. The fundamental error component induces permanent price changes, and is related to the fundamental volatility that consists of random price changes that do not revert. The transitory component is related to the transitory volatility that is taken as an implicit transaction cost and a measure of the market quality, (Hasbrouck, 1993). As such, it is important and relevant for market regulators to be able to distinguish between the fundamental and the transitory volatility when addressing issues related to excess volatility. A standardized measure of market quality can be derived as (Ben Sita and Westerholm, 2006):

$$\rho = \frac{-\sigma_n^2 + (2(\phi_1\phi_2 + \phi_2\phi_3 + \phi_1\phi_3) + \phi_1^2)\rho_q}{(\phi_1 + \phi_2 + \phi_3)^2 + \rho_q\theta_1^2 + \sigma_\varepsilon^2 + 2\sigma_n^2} \qquad (3)$$

where ρ is the ratio of the first order covariance to the variance. The larger the absolute value of the ratio, ignoring intensity effects on the price process, the better a market's quality of trading, since a lower coefficient is an indication that the duration and volume variables either increase

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the covariance bias or decrease the variance in the trading process. A smaller variance or greater covariance suggests greater pricing efficiency and thus improved market quality.

3.3 The ACD Model

This analysis employs an Autoregressive Conditional Duration (ACD) approach that directly models the time between events to explicitly model time, a formulation particularly suited to irregularly spaced data. In this framework, the time elapsed between two market events, (for example, a trade, a specific type of trade such as a large buy, a revision of the bid-ask quote, and so on) conveys meaningful information and is the object of modeling. The ACD is a type of dependent point process particularly suited for modeling characteristics of duration series, such as clustering and overdispersion. This parameterization is most easily expressed in terms of the waiting time between events.

Let $x_t = T_t - T_{t-1}$ be the interval of time between event arrivals, our measure of duration. The distribution of the duration is conditioned directly upon past duration. According to Engle and Russell (2004), the ACD model is specified by taking the expectation of duration given past arrival times, ψ_t, such that:[8]

$$E(x_t | x_{t-1}, x_{t-2}, \ldots, x_1) = \psi_t(x_{t-1}, x_{t-2}, \ldots, x_1) = \psi_t \tag{4}$$

and,

$$x_t = \psi_t \varepsilon_t \tag{5}$$

where $\varepsilon_t \sim$ i.i.d. with density $p(\varepsilon;\varphi)$. Engle and Russell (1998) suggest and subsequently apply a linear parameterization for the expectation which is given by:

$$\psi_t = \omega + \sum_{j=1}^{p} \alpha_j x_{t-j} + \sum_{j=1}^{q} \beta_j \psi_{t-j} \tag{6}$$

since the conditional expectation of the duration depends on *p* lags of the duration and *q* lags of the expected duration, this is termed an ACD(*p*,*q*) model. The ACD model is utilised as a model

[8] We refer the interested reader to Engle and Russell (1998) for a more detailed discussion.

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for intertemporally correlated event arrival times and seeks to capture trading time (duration) clustering. In analogous fashion, the Autoregressive Conditional Volume (ACV) trade volume clustering can be captured using an Autoregressive Conditional Volume (ACV) model, ACV(*p*,*q*), specified as:

$$z_t = \tau + \sum_{j=1}^{p} \alpha_j w_{t-j} + \sum_{j=1}^{q} \beta_j z_{t-j} \tag{7}$$

where $w_t = v_t - v_{t-1}$. Both the ACD(*p*,*q*) model of equation (6) and the ACV(*p*,*q*) model of equation (7) exhibit strong similarities to the (G)ARCH(*p*,*q*) models of Engle (1982) and Bollerslev (1986) and can be modeled as such. Further the models can be augmented with extra explanatory variables. We augment the ACD (ACV) with past changes in volume (duration) to examine and to explicitly account for the inter-relationship between volume and duration. We theorize that the relationship between duration and volume has a significant influence on price changes. In particular, we hypothesize that price effects increases for larger volumes traded at longer durations. To test this hypothesis, we use the estimates of the expected duration, ψ_t, from the A-ACD model, as well as the estimate of expected volume, z_t, from the A-ACV model.

3.4 Alternative Empirical Models

In order to provide a robustness check and cross-validate our findings in the context of existing literature, we also examine a set of closely related alternative models for the estimation of the implied bid-ask spread. These are the models proposed by Madhavan et al. (1997) (equation 8a), Dufour and Engle (2000) (equation 8b), De Jong et al. (1996) (equation 8c) and Ben Sita and Westerholm (2006) (equation 8d):

$$r_t = \theta(q_t - \rho_q q_{t-1}) + \phi(q_t - q_{t-1}) + u_{1t} \tag{8a}$$

$$r_t = (\gamma_0 q_t + \gamma_1 q_{t-1}) + [\delta_0 \ln(x_t) q_t + \delta_1 \ln(x_{t-1}) q_{t-1}] + u_{2t} \tag{8b}$$

$$r_t = a_0 + (R_0 \Delta q_t + R_1 \Delta q_t z_t) + (e_0 q_{t-1} + e_1 q_t z_{t-1}) + u_{3t} \tag{8c}$$

$$r_t = \lfloor \phi_1 \Delta q_t + (\phi_2 \psi_t + \rho_q \theta_1 C_t) q_t \rfloor + (\varepsilon_t + (n_t - n_{t-1})) \tag{8d}$$

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where u_{1t}, u_{2t}, and u_{3t} are random error terms. In equation (8a) the order processing cost is captured by ϕ and the adverse selection cost by θ. In equation (8b) the order processing costs are given by $(\gamma_0 - \gamma_1)q$ and the adverse selection costs by $(\delta_0 - \delta_1)\ln(x)q$ while in equation (8c) the order processing cost component is $R_0 - e_0 - e_1\alpha + (R_1 - 0.5e_1)z$, where α is the median of transaction size distribution (divided by the natural logarithm of 2), and the adverse selection component is given by $e_0 + e_1\alpha - 0.5e_1 z$. This model differs from the other two, since price revisions in this model are associated with changes in the expectations of the true value of the stock, given the size of the transaction. In equation 8(d) the Ben Sita and Westerholm (2006) specification extends the Madavan et al. (1997) model by adding duration measures to the structural model to measure the contribution of durations to the bid-ask spread in a limit order book market. In this context, the order processing cost is captured by ϕ_1 and the adverse selection cost is given by θ_1. We estimate equations (8a) and (8d) using a maximum likelihood estimator (MLE). This method is preferred to the generalized method of moments (GMM), simply on the basis that the MLE provides more efficient (robust) coefficients and corrects for any misspecification in the model. Further, by using MLE, we eliminate any potential imprecision that may arise in the results from over- or under- identification in the GMM. Equations (8b) and (8c) are estimated using an OLS estimator with Heteroskedasticity and Autocorrelation consistent Covariance (HAC) robust errors using the Newey-West estimator.

4 Empirical Analysis

4.1 Data

The Mexican Derivatives Exchange (MexDer) commenced operations on December 15, 1998, trading a Mexican Peso-US Dollar futures contracts. Trading on the stock index (IPC) began on April, 15 1999, and in interest rate futures one month later. The market operated via open outcry until May 8, 2000 when electronic trading was introduced. Trading on the MexDer is now conducted through a completely automated system, such that prices are revealed to the general public both domestically and globally. Market makers trade in a limit order book market

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and maintain bid and ask quotes for futures contracts for which they are registered, in order to promote their trading.

We examine trading on the MexDer using futures data spanning the period from August 28, 2000 to October 13, 2006. Specifically, we use tick by tick data for the 28-day equilibrium interbank interest rate futures contract (Tasa de Interes Interbancaria de Equilibrio or TIIE28), the most actively traded futures contract listed on the MexDer. The trading session for this contract runs from 7:30am to 2:15pm Mexico City time. The contract is based on the short-term (28-day) Mexican interbank interest rates, has a face value of 100,000 Mexican pesos with a yield expressed in annualized percentile terms. A 28-day TIIE contract is cash settled based on the 28-day TIIE calculated by the Banco de Mexico based on quotations submitted daily by full-service banks using a mechanism designed to reflect conditions in the Mexican Peso Money Market. MexDer lists TIIE futures contracts with maturity dates on a monthly basis for up to the next ten years. The data includes comprehensive information relating to all transactions, maturities, trade prices, opening prices and trading volume for 2,220 trading days giving rise to 88,562 recorded transactions. However, these transactions do not equal the 88,562 information ticks since some transactions in close proximity to each other are redundant.

To prepare the data for the analysis, we drop any transactions that are recorded as occurring either before the official opening time, 7:30am, or after the designated closing time 2:15pm. To obtain active price movements, we also thin the trade price process in a similar fashion to Ben Sita and Westerholm (2006). This thinning process consists of ignoring a price at time $t-1$ if this price equals the price at time t and in such a case adding the trading volume of the price at time $t-1$ to that at time t. We then compute the duration between trades, treating the overnight period as if it did not exist, for example, implying the time elapsing between 13:59:30 of day t-1 and 7:30:10 of day t is only 40 seconds. We adopt this strategy due to the infrequent trading nature of the futures contracts. Manganelli (2005) argues that eliminating the duration of the overnight period causes the loss of important information for futures trading. The underlying assumption here is that the information flow to the market is continuous. By eliminating the duration of the overnight period when there is no trade, we remove the suggestion that there is no price relevant news, which is otherwise implied by including the unadjusted overnight duration. Additionally, we eliminate all transaction data with a zero duration. We treat these transactions

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as one single transaction, summing up all relevant volumes. Taking these adjustments into consideration, the sample size is reduced from 88,562 to 69,981 information ticks.

Furthermore, we also adjust for the time of day effects which potentially influence both duration and volume. There is a consensus in the literature that both duration and volume exhibit a typical pattern over the course of the trading day, with very high trading activity occurring both at the beginning and at the end of a typical session. In order to remove this seasonal feature from the data, the duration and volume series were diurnally adjusted in the manner suggested by Engle (2000). To adjust both the trade duration and volume for the time of day effects, we regress duration and volume on a piecewise linear function of the daily trading time, namely:

$$\tilde{x}_t = x_t / E\left[x_t \middle| f(t)\right] \tag{9}$$

$$\tilde{w}_t = w_t / E\left[w_t \middle| f(t)\right] \tag{10}$$

where

$$f(t) = \beta_0 + \sum_{j=1}^{4} \beta_j (t - k_j)$$

for k_j fixed at 27000 (7:30am), 34200 (9:30am), 41400 (11:30am) and 48600 (1:30pm) seconds.[9]

4.2 Descriptive Statistics

In Table 1 we report summary statistics for the variables used in the analysis. We note that the returns (r_i) exhibit negative skewness and leptokurtosis, typical characteristics of financial returns data. Negative skewness is an indication that the probability of observing a large negative jump in the interest rate futures price exceeds the corresponding positive jump probability during the sample period. The returns indicate that there is both clustering and overdispersion in the data (the standard deviation is larger than the mean). The adjusted variables ($\tilde{x}_i ; \tilde{w}_i$) are close to their expected values of unity. Dealing with adjusted variables allows us to investigate the effects of excess dispersion on the returns, enabling us to investigate the extent to

[9] We note that β_k are consistent (but inefficient) OLS estimates, as we regress the respective trade duration and trade volumes on *f(t)*.

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which incorporating adjusted duration and volume might enhance the analysis of Madhavan et al. (1997) that implicitly assumes duration and volume effects equal 1. Neither the mean nor the median of the trade indicator variable, (q_i), gives any indication that this variable is zero as expected under the null hypothesis that a trade at bid (ask) is immediately followed by a trade at ask (bid). The derived variable (C_i) which measures the covariance of the standardized residual of duration, is an attempt to capture the forces that serve to drive prices away from their fundamental values. The summary statistics of table 1 indicate that the average duration between successive events for the interest rate futures during exchange opening times is approximately 1 hour and 33 minutes. The average trading volume corresponding to each time stamp is 1.7 million Mexican pesos, with each contract traded corresponding to 100,000 Mexican pesos.

4.3 Estimation Results

Table 2 reports the coefficients for the ACD(1,1) model and the volume augmented-ACD(1,1) model. The results of the ACD(1,1) model in table 2a indicate that the model captures trading momentum. The basic ACD model appears to capture the dynamics of the duration process reasonably well, signifying that the duration process is significantly persistent. The coefficients are all positive and significant, implying that long (short) durations tend to be followed by long (short) durations. When augmented to include volume in table 2b, the coefficients of the A-ACD(1,1) model again reveal a persistent duration process. The coefficient for the volume variable, however, indicates that there is a significantly negative relationship between duration and trading volume. The implication is that increased trading volume results in higher price volatility. These results may be indicating the prominence and influence of informed traders on price formation.

The estimated autoregressive coefficients of the ACV(1,1) model and the duration augmented-ACV(1,1) model, are reported in table 3. Table 3a reveals that all the coefficients are highly significant. Our results show that changes in volume exhibit persistence, providing evidence of trading volume clustering. This is consistent with Admati and Pfleiderer (1988) who demonstrate that, in equilibrium, all discretionary liquidity traders choose to trade together generating clustering of trading volume. A potential underlying intuition for clustering in the trading volume revolves around the behavior of discretionary liquidity traders. The decision of discretionary liquidity traders to postpone their trading until an announcement has been released

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will make it easier for the market to infer the informed agent's reasons for trading. The coefficients of the ACV(1,1) model are all positive and significant, indicating that large transaction size increases the expected transaction size of the next trade. The coefficients of the augmented ACV(1,1) model (table 3, panel b) tentatively suggest a negative relationship between trading volume and duration, implying that shorter durations result in higher trading volume volatility. This corroborates the theoretical predictions in the market microstructure literature suggesting that durations are informative about intra-day trading volatility, specifically the shorter the duration, the higher the trading intensity.

Table 4, panel A reports the coefficients obtained from the maximum likelihood estimates of the model in equation (2). The results indicate a significant relationship between both volume and duration on pricing effects. However, contrary to the prediction that the price impact increases with transaction size and/or trade volume, the ϕ_3 coefficient is negative and significant, indicating that price impact decreases with volume.[10] The proposition in the existing literature that prices increases when traders observe short durations is predicated upon the link between short durations and good news and the possible existence of short-sales constraints precluding the ability to sell an asset when encountering bad news.[11] Here, the coefficient ϕ_2 captures the relationship between duration and price changes and is positive and significant, indicating that greater price increases are associated with longer duration. The positive coefficient on the duration variable indicates that transactions arriving after a long time have a larger price impact than transaction arriving after a previous trade. The intuition is that when trades are further apart, it is more difficult to find suppliers/traders willing to take the opposite side of a transaction, and so trades have a higher impact on prices.

However, one must view this apparent inconsistency in findings in the appropriate market context. For interest rate futures unlike assets such as equities there is no a priori reason to believe that expected changes in the underlying in any given direction represent "better" news than those in the opposite direction. Moreover, unlike cash market equity trades, in a futures market there is no asymmetry in terms of a trader's ability to assume a long (buy) or short (sell) position, so short-sales constraints on trading are irrelevant. The ϕ_2 coefficient is the largest in

[10] De Jong et al. (1996). Also note that this finding is dependant upon the precise specification of the model and the use of mid-quotes versus transaction prices.
[11] Dufour and Engle (2000).

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absolute magnitude, implying that the price impact originates mainly from the expected duration of trade, with longer durations generating greater price effects. The expected duration innovation coefficient, θ_1, is, however, negatively related to returns. Despite the liquidity (ϕ_2) and information (θ_1) coefficients have contrasting effects on returns, the dominance of the liquidity component suggests liquidity traders dictate the time to trade. This interpretation is consistent with the results of the ACV model.

In panel B of table 4 we report the results of the model of equation (2) using the estimates from the A-ACD and A-ACV models. We maintain that the relationship between volume and duration has significant influence on price effects. The results of the model appear to confirm this, since the coefficients on duration and volume are much larger than in the model using estimates from the ACD and ACV models. Further, we highlight that both the transitory and permanent duration components are positive and significant. Importantly, the coefficient on the permanent duration term (C_t) is now positive and significant. The reversal in sign and increase in size of the coefficient confirms the significant influence of the relationship between volume and duration and implies that price effects are indeed larger for volumes traded at longer durations.

5 Sensitivity Analysis

5.1 Alternative Empirical Models

Table 5 presents the empirical results of the alternative models. Specifically, panel A reports the results of the Madhavan et al. (1997) model which uses only the trade indicator variable to split the price effects into transitory and permanent effects. In panel B we present the results of the Dufour and Engle (2000) model which uses the duration variable to investigate whether it can capture trading behavior. Panel C reports the results of the De Jong et al. (1996) specification which includes trade size in the structural model. Finally, panel D gives the Ben Sita and Westerholm (2006) specification which include both trade indicator and duration variables to decompose the price effects into transitory and permanent effects. The general results from these models indicate that, with the exception of Ben Sita and Westerholm's (2006) model, order handling cost is the dominant component influencing price effects.

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The implied bid-ask spread and the components are reported in table 6. The results show that the implied bid-ask spread is 2.1014 and is based on trade prices in the TIIE. The proportion of the adverse selection cost in the spread is 22.74 per cent, while liquidity costs account for 73.74 per cent of the implied spread. The comparative evidence from the other models reveal that our present model produces the largest implied bid-ask spread with the smallest spread of 0.1388 obtained in the De Jong et al (1996) model. The liquidity component of the implied spread in the Ben Sita and Westerholm (2006) model is 64.33 per cent, 9.4 percentage points smaller than that of the current model and is partly attributable to the exclusion of the volume variable. In essence, the implied bid-ask spread lies between 0.1388 for the lowest case based on the De Jong et al (1996) model and 2.1014 the largest according to our model. A cursory look at the results indicates that the implied bid-ask spread increases with the inclusion of more trade process variables. The theoretical motivation for this is not known and would require further investigation.

Analogous to Madhavan et al (1997) model's decomposition of price volatility, we decompose our model's estimation of price volatility accordingly. The results are reported in Table 7. From equation (2) we derive price volatility as

$$\text{var}[r_t] = (\phi_1 + \phi_2 + \phi_3)^2 + \rho_q \theta_1^2 + \sigma_\varepsilon^2 + 2\sigma_n^2 \qquad (11)$$

where σ_ε^2 reflects the variance in public news shocks. The remaining components such as price discreteness ($2\sigma_n^2$), adverse selection costs ($\rho_q \theta_1^2$), order handling costs (ϕ_1^2), liquidity costs ($\phi_2^2 + \phi_3^2 + 2\phi_2\phi_3$), and an interaction cost ($2\phi_1\phi_2 + 2\phi_1\phi_3$) constitute market microstructure noise. Intuitively, the fraction of variance attributable to trading frictions (including price discreteness) is π_1, while the fraction of volatility attributable to pricing errors is π_2, where

$$\pi_1 = \frac{2\sigma_n^2 + \rho_q \theta_1^2 + (\phi_1 + \phi_2 + \phi_3)^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \rho_q \theta_1^2 + (\phi_1 + \phi_2 + \phi_3)^2} \qquad (12)$$

and

$$\pi_2 = \frac{2\sigma_n^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \rho_q \theta_1^2 + (\phi_1 + \phi_2 + \phi_3)^2} \qquad (13)$$

The estimates of our model produce a price risk of 0.0247, comparable to the price volatility of 0.0188 from the Ben Sita and Westerholm (2006) model. These two values are approximately half of the price risk of 0.0449 derived in the Madhavan (1997) model. This

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striking difference indicates that, ceteris paribus, accounting for duration and volume decreases price volatility. One explanation maybe that better hedging strategies are devised when more market microstructure variables are taken into consideration. This is a salient feature that should be considered by traders and investors. Interestingly, the proportion of variance attributable to pricing errors, π_2, is much smaller for the Madhavan (1997) model, 0.00125, compared to our estimate of 0.0226. The fraction of variance attributable to trading frictions, π_1, is 95.41 per cent and differs slightly from the respective estimates of 97.44 per cent and 94.27 per cent for Madhavan's and Ben Sita and Westerholm's models. This result accords with the perception that price volatility in the TIIE28 contract on the MexDer is largely the result of market microstructure noise.

Overall, the results are somewhat contrasting with those from our volume augmented duration structural specification. In particular in these models the order handling cost component is generally more dominant in generating pricing impact, whereas in our present model duration is the dominant factor. Nonetheless, perhaps the most salient point regarding the results of all models is that the theoretical predictions of the microstructure models appear reliant on the model specification and the underlying nature of the markets.

5.2 Nearby Contracts

As a further robustness check, the empirical estimation is executed on data taken exclusively from the nearby contracts (those closest to maturity), the rationale being that they customarily exhibit the highest liquidity and convey the important information on investor trading patterns. Nearby contracts defined in this market context are those trading in the month prior to their maturity: we do not include observations on contracts trading in the maturity month, simply to avoid biases caused by unusual market activity near maturity as traders unwind or rollover positions. The data pertaining to nearby contracts are the tick by tick trade prices for contracts trading in the month prior to maturity. The sample of tick by tick trades on the TIIE28 contracts on the MexDer consists of 11,687 trades in nearby contracts covering the period August 28, 2000 to October 13, 2006. To analyze the price impact of volume, duration and orderflow we estimate the empirical models outlined above using the nearby contracts.

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The empirical results of the structural model outlined in table 8 indicate that the results are consistent with those from the full sample model.[12] In particular, the coefficient for expected duration (ϕ_2) is positive and significant at the 10 per cent level and volume has a negative effect on prices. The results indicate that there is no significant change in the trading impact of trade process variables when contracts are close to maturity. This is further supported by the results of the alternative empirical microstructure models reported in table 9.

6 Conclusions

This paper examines the pricing impact of duration in the context of a structural microstructure model of price discovery using data from the emerging futures market of Mexico, MexDer. We incorporate both elapsed transaction time and trade volume in the model in order to investigate their effects on price changes. In their empirical study, Dufour and Engle (2000) document that long duration has a lower impact on price than short duration. Extending Dufour and Engle (2000) by decomposing the duration effect into liquidity and information effects, Ben Sita and Westerholm (2006) find evidence that both components exert a positive influence on price changes.

We examine six years of tick by tick data on the 28-day equilibrium interbank interest rate futures contract (TIIE28) on MexDer. In November 2005 this contract accounted for 94.5% of total MexDer trading volume and was one of the most actively traded futures contracts world-wide. Our findings involving transaction duration show that the transient and permanent components of duration exert positive and negative influences on price changes, respectively. Overall, the net effect is similar to the evidence provided by Ben Sita and Westerholm (2006). In contrast to Dufour and Engle (2000), we find that long duration has a greater impact on price changes than short duration. Our analysis of trading volume suggests that price changes tend to decrease when there is an increase in trade volume. Additionally, increases in order flow exert a positive influence on price changes, albeit to a lesser extent than the influence of duration. Viewed collectively, this evidence suggests that managing time and trading volume are important aspects of optimum trading strategy in the Mexican futures market. Further, the results of this paper suggest that trading constraints such as position limits, or short sales constraints

[12] For space considerations we only report the final model results using nearby contracts. Full results are available from the corresponding author on request.

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dictated by the specific institutional market context in which trading occurs may be relevant in determining the pricing impact of trading process variables.

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References

Admati, A., Pfleiderer, P., 1988. A theory of intraday patterns: volume and price volatility. Review of Financial Studies 1, 3-40.

Back, K., Baruch, S., 2004. Information in securities market: Kyle meets Glosten and Milgrom. Econometrica 2, 433-467.

Ben Sita, B., Westerholm, P.J., 2006. The role of time in price discovery: Ultra-high frequency trading in a limit order book market. Discussion paper, University of Sydney.

Bollerslev, T., 1986. Generalized Autoregressive Conditional Heteroskedasticity. Journal of Econometrics 31, 307-327.

Chung, K.H., Li, M., McInish, T., 2005, Information based trading, price impact of trades, and trade autocorrelation, Journal of Banking and Finance 29, 1645-1669.

Clark, P., 1973. A subordinated stochastic process model with finite variance for speculative prices. Econometrica 41, 135-156.

Diamond, D., Verrecchia, R.E., 1987. Constraints on short-selling and asset price adjustments to private information. Journal of Financial Economics 18, 277-311.

DeJong, F., Nijman, T., Roëll, A., 1996. Price effects of trading and components of the bid-ask spread on the Paris Bourse. Journal of Empirical Finance 3, 193-213.

Dufour, A., Engle, R.F., 2000. Time and the Price Impact of a Trade. Journal of Finance 55, 2467-2498.

Easley, D., O'Hara, M., 1987. Price, trade size, and information in securities markets. Journal of Financial Economics 19, 69-90.

Easley, D., O'Hara, M., 1992. Time and the process of security price adjustment. Journal of Finance 47, 557-605.

Easley, D., Kiefer, M.N., O'Hara, M., 1997. The information content of the trading process. Journal of Empirical Finance 4, 159-186.

Engle, R.F., 1982. Autoregressive Conditional Heteroskedasticity with estimates of the variance of U.K. inflation. Econometrica 61, 909-927.

Engle, R.F., 1996. The econometrics of ultra-high frequency data. Discussion paper 96-15, Department of Economics, University of California, San Diego.

Engle, R.F., 2000. The econometrics of ultra-high frequency data. Econometrica 68, 1-22.

Electronic copy available at: https://ssrn.com/abstract=1025241

Engle, R.F., Lange, J., 2001. Predicting VNET: A model of the dynamics of market depth. Journal of Financial Markets 4, 113-142.

Engle, R.F., Lunde, A., 2003. Trades and quotes: A bivariate point process. Journal of Financial Econometrics 1, 159-188.

Engle, R.F., Russell, J.R., 1997. Forecasting the frequency of changes in quoted foreign exchange prices with the autoregressive conditional duration model. Journal of Empirical Finance 4, 187-212.

Engle, R.F., Russell, J.R., 1998. Autoregressive conditional duration: a new model for irregularly spaced transactions data. Econometrica 66, 1127-1162.

Engle, R.F., Russell, J.R., 2004. Analysis of high frequency data. Discussion paper 04-10, Department of Economics, University of California, San Diego.

Ghysels, E., Gouriéroux, C., Jasiak, J., 2004. Stochastic volatility duration models. Journal of Econometrics 119, 413-433.

Ghysels, E., Jasiak, J., 1998. GARCH for irregularly spaced financial data: The ACD-GARCH model. Studies on Nonlinear Dynamics and Econometrics 2, 133-149.

Glosten, L., Milgrom, P., 1985. Bid, ask, and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics 14, 71-100.

Glosten, L., Harris, L., 1988. Estimating the components of the bid-ask spread. Journal of Financial Economics 21, 123-142.

Gouriéroux, C., Jasiak, J., LeFol, G., 1999. Intraday trading activity. Journal of Financial Markets 2, 193-226.

Grammig, J., Wellner, M., 2002. Modeling the interdependence of volatility and inter-duration processes. Journal of Econometrics 106, 369-400.

Hasbrouck, J., 1991. Measuring the information content of stock trades. Journal of Finance 46, 179-207.

Hasbrouck, J., 1993. Assessing the quality of a security market: A new approach to transaction-cost measurement. Review of Financial Studies 6, 191-212.

Holder, M.E., Qi, M., Sinha, L., 2004. The Impact of Time Duration between trades on the price of Treasury Note Futures contracts. Journal of Futures Markets 24(10), 965-980.

Huang, R.D., Stoll, H.R., 1997. The components of the bid-ask spread: A general approach, Review of Financial Studies 10, 995-1034.

Electronic copy available at: https://ssrn.com/abstract=1025241

Ho, T.S.Y., Stoll, H.R., 1983. The Dynamics of dealer markets under competition. Journal of Finance 38, 1053-1075.

Holz, R., 2007. Nearly 10 billion contracts traded worldwide in 10 months. Futures Industry Magazine, January-February, 10-13.

Kyle, A.S., 1985. Continuous auctions and insider trading. Econometrica 53, 1315-1335.

Locke, P., Onayev, Z., 2005. Trade Duration: Information and Trade Disposition. The Financial Review 40, 113-129.

Madhavan, A., Richardson, M., Roomans, M., 1997. Why do security prices change? A transaction-level analysis of NYSE stocks. Review of Financial Studies 10, 1035-1064.

Manganelli, S., 2005. Duration, volume, and the price impact of trades. Journal of Financial Markets 8, 377-399.

O'Hara, M., 1995. Market Microstructure Theory. Basil Blackwell.

O'Hara, M., 2003. Presidential address: Liquidity and price discovery. Journal of Finance 58, 1335-1354.

O'Hara, M., Oldfield, G.S., 1986. The Microeconomics of market making. Journal of Financial and Quantitative Analysis 21, 361-376.

Renault, E., Werker, B.J.M., 2002. Stochastic volatility with transaction time risk. Discussion paper, University of Montreal.

Roll, R., 1984. A simple implicit measure of effective bid-ask spread in an efficient market. Journal of Finance 39, 1127-1139.

Zhong, M., Darrat, A.F., Otero, R., 2004. Price discovery and volatility spillovers in index futures markets: Some evidence from Mexico. Journal of Banking and Finance 28, 3037-3054.

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Table 1 : Summary Statistics

This table presents descriptive statistics for the analysed series based on tick-by-tick data for the TIIE28, 28-day interbank interest rate futures contract on MexDer. The sample consists of 69,981 information ticks between August 28, 2000 and October 13, 2006. r_i is the return, x_i is duration, $\tilde{x}_i$ is adjusted duration, w_i is the change in trading volume, $\tilde{w}_i$ is the adjusted change in trading volume, ψ_i is the conditional duration from an ACD(1,1) model, z_i is conditional change in trading volume from an ACV(1,1) model, q_i is the trade indicator and C_i is the signed duration innovation.

	Mean	**Median**	**Std Dev**	**Skewness**	**Kurtosis**	**Minimum**	**Maximum**
r_i	-0.0347	0.0000	1.1984	-0.8323	50.1478	-25.2644	23.6533
x_i	8.4783	8.0161	2.8096	-0.1444	2.0298	4.6052	14.3433
$\tilde{x}_i$	1.0002	1.0004	0.0336	-128.0778	29074.22	-6.0735	3.5986
w_i	7.4780	7.6009	1.0200	-1.8300	17.9973	0.0000	12.9575
$\tilde{w}_i$	0.9954	1.0003	0.0677	-14.6110	214.7299	0.0000	1.2499
ψ_i	1.0002	1.0002	0.0007	128.0758	29073.61	0.9476	1.1433
z_i	0.9954	0.9968	0.0199	-14.6109	214.7268	0.7027	1.0702
q_i	-0.0063	-1.0000	1.0000	0.0126	1.0002	-1.0000	1.0000
C_i	0.0002	-0.0002	0.0190	86.6003	9627.237	-0.2479	2.4773

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Table 2 : Results of Autoregressive Conditional Duration models

This table reports the coefficient estimates and corresponding t-statistics from two Autogressive Conditional Duration models. Panel A reports the ACD(1,1) model:

$$\psi_t = \omega + \alpha x_{t-1} + \beta \psi_{t-1}$$

Panel B reports the Augmented-ACD(1,1) model:

$$\psi_t = \omega + \alpha x_{t-1} + \beta \psi_{t-1} + \delta w_{t-1}$$

A: ACD(1,1) model		
	Coefficient	**t-statistic**
ω	0.0031	6.4524
α	0.0400	22.5914
β	0.9530	371.6673

B: A-ACD(1,1) model		
	Coefficient	**t-statistic**
ω	0.0031	6.4635
α	0.0398	22.6462
β	0.9532	374.9224
δ	-0.2874	-8.6460

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Table 3 : Results of Autoregressive Conditional Volume models

This table reports the coefficient estimates and corresponding t-statistics from two Autoregressive Conditional Volume models. Panel A reports the ACV(1,1) model:

$$z_t = \tau + \alpha w_{t-1} + \beta z_{t-1}$$

Panel B reports the Augmented-ACV(1,1) model:

$$z_t = \tau + \alpha w_{t-1} + \beta z_{t-1} + \theta x_{t-1}$$

A: ACV(1,1) model		
	Coefficient	**t-statistic**
τ	0.0010	62.9227
α	0.0421	97.6075
β	0.9466	2539.919

B: A-ACV(1,1) model		
	Coefficient	**t-statistic**
τ	0.0010	62.8901
α	0.0421	97.5914
β	0.9467	2542.334
θ	-0.0613	-1.4790

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Table 4 : Estimation Results of the Trade model

This table reports the coefficient estimates and corresponding t-statistics of the estimated trade equation:

$$r_t = \lfloor \phi_1 \Delta q_t + (\phi_2 \psi_t + \rho_q \theta_1 C_t + \phi_3 z_t) q_t \rfloor + (\varepsilon_t + n_t - n_{t-1})$$

Estimation is based on tick-by-tick data for the TIIE28, 28-day interbank interest rate futures contract on MexDer using a sample consisting of 69,981 information ticks between August 28, 2000 and October 13, 2006.

A: Using estimates from ACD(1,1) and ACV(1,1) models		
	Coefficient	**t-statistic**
ϕ_1	0.0370	8.1807
ϕ_2	0.4448	4.7969
ϕ_3	-0.3300	-3.5241
θ_1	-0.2389	-11.7153
B: Using estimates from A-ACD(1,1) and A-ACV(1,1) models		
	Coefficient	**t-statistic**
ϕ_1	0.0522	11.5217
ϕ_2	1.0258	10.6348
ϕ_3	-0.8854	-9.0935
θ_1	0.7690	11.6801

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Table 5 : Estimation Results of Alternative Empirical Models

This table reports the coefficient estimates and corresponding t-statistics from the estimation of four alternative empirical models. Panel A reports the Madahavan et al. (1997) model:

$$r_t = \theta\left(q_t - \rho_q q_{t-1}\right) + \phi\left(q_t - q_{t-1}\right) + u_t$$

Panel B reports the Dufour and Engle (2000) model:

$$r_t = \left(\gamma_0 q_t - \gamma_1 q_{t-1}\right) + \left[\delta_0 \ln(x_t) q_t + \delta_1 \ln(x_{t-1}) q_{t-1}\right] + u_t$$

Panel C reports the DeJong et al. (1996) model:

$$r_t = a_0 + \left(R_0 \Delta q_t + R_1 \Delta q_t z_t\right) + \left(e_0 q_{t-1} + e_1 q_t z_{t-1}\right) + u_t$$

Panel D reports the Ben Sita and Westerholm (2006) model:

$$r_t = \left[\phi_1 \Delta q_t + \left(\phi_2 \psi_t + \rho_q \theta_1 C_t\right) q_t\right] + \left(\varepsilon_t + n_t - n_{t-1}\right)$$

Estimation is based on tick-by-tick data for the TIIE28, 28-day interbank interest rate futures contract on MexDer using a sample consisting of 69,981 information ticks between August 28, 2000 and October 13, 2006.

A: Madahavan et al. (1997) model

	Coefficient	t-statistic
θ	0.0187	19.9909
ϕ	0.1374	34.8376

B: Dufour and Engle (2000) model

	Coefficient	t-statistic
γ_0	0.1561	34.4212
γ_1	-0.03427	-7.5591
δ_0	0.0000	0.2616
δ_1	-0.0001	-0.5779

C: DeJong et al. (1996) model

	Coefficient	t-statistic
a_0	-0.0339	-7.5462
R_0	0.1970	10.5701
R_1	-0.0058	-0.6470
e_0	0.1566	9.2008
e_1	-0.0354	-2.2170

D: Ben Sita and Westerholm (2006) model

	Coefficient	t-statistic
ϕ_1	0.0345	7.625
ϕ_2	0.1221	20.027
θ_1	0.0332	0.3105

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Table 6 : Estimation Results of the implied bid-ask spread

This table reports the estimates of the implied bid-ask spread from each of the respective models, the implied spread in our trade model is $S = 2(\theta_1 + \phi_1 + \phi_2 + \phi_3)$. The proportion of adverse selection, order handling and liquidity costs in spread are given by $\vartheta_1 = \frac{\theta_1}{\theta_1+\phi_1+\phi_2+\phi_3}$, $\vartheta_2 = \frac{\phi_1}{\theta_1+\phi_1+\phi_2+\phi_3}$ and $\vartheta_3 = \frac{\phi_2+\phi_3}{\theta_1+\phi_1+\phi_2+\phi_3}$. In the Ben Sita and Westerholm (2006) model (BS & W) the implied spread is $S = 2(\theta + \phi_1 + \phi_2)$ with the proportion of adverse selection cost, order handling and liquidity costs given by $\vartheta_1 = \frac{\theta}{\theta+\phi_1+\phi_2}$, $\vartheta_2 = \frac{\phi_1}{\theta_1+\phi_1+\phi_2}$ and $\vartheta_3 = \frac{\phi_2}{\theta+\phi_1+\phi_2}$. In Madahavan et al. (1997) model the implied spread is $S = 2(\theta + \phi)$ while in the Dufour and Engle (2000) model (D&F) it is given by $S = 2(\varphi + \vartheta)$, where $\varphi = \mid (\gamma_0 - \gamma_1) \mid q$ and $\vartheta = \mid \delta_0 - \delta_1 \mid \ln(\tilde{x}_i)$, and the proportion of adverse selection cost is $\vartheta_1 = \frac{\vartheta}{\vartheta+\varphi}$. In the DeJong et al. (1996) model, the implied spread is calculated as $S = 2(\varphi + \vartheta)$ where $\varphi = R_0 - e_0 - e_1\alpha + (R_1 - 0.5e_1)\tilde{z}_i$ and $\vartheta = e_0 + e_1\alpha - 0.5e_1\tilde{z}_i$ with the proportion of adverse selection cost given by $\vartheta_1 = \frac{\vartheta}{\vartheta+\varphi}$.

Estimates of the implied bid-ask spread and the components

	Our Model	BS & W	Madhavan et al.	D & E	DeJong et al.
S	2.1014	0.3796	0.3122	0.3807	0.1388
ϑ_1	22.74	17.49	11.98	0	61.29
ϑ_2	3.52	18.18	88.02	100.0	38.71
ϑ_3	73.74	64.33			

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Table 7 : Estimation Results of Price Volatility

This table reports the estimates of price volatility and its components for the respective models. For our trade model, price volatility is calculated as $\text{var}[\Delta p_t] = [\phi_1 + \phi_2 + \phi_3]^2 + \rho_q + \theta_1^2 + \sigma_\varepsilon^2 + 2\sigma_n^2$, with the fraction of the variance attributable to trading frictions, π_1, and the fraction of variance attributable to pricing errors, π_2, given by $\pi_1 = \frac{2\sigma_n^2 + \rho_q\theta_1^2 + (\phi_1+\phi_2+\phi_3)^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \rho_q\theta_1^2 + (\phi_1+\phi_2+\phi_3)^2}$ and $\pi_2 = \frac{2\sigma_n^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \rho_q\theta_1^2 + (\phi_1+\phi_2+\phi_3)^2}$. For the Ben Sita and Westerholm (2006) model, price volatility is calculated as $\text{var}[\Delta p_t] = \phi_1^2 + \phi_2^2 + \theta_1^2 + 2\theta_1\phi_2\rho_q + \sigma_\varepsilon^2 + 2\sigma_n^2$ and the fraction of the variance attributable to trading frictions, π_1, is $\pi_1 = \frac{2\sigma_n^2 + \phi_1^2 + \phi_2^2 + \theta_1^2 + 2\theta_1\phi_2\rho_q}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \theta_1^2 + \phi_1^2 + \phi_2^2 + 2\theta_1\phi_2\rho_q}$ and the fraction of variance attributable to pricing errors, π_2, is given by $\pi_2 = \frac{2\sigma_n^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + \theta_1^2 + \phi_1^2 + \phi_2^2 + 2\theta_1\phi_2\rho_q}$. For the Madhavan et al. (1997) model, price volatility is calculated as $\text{var}[\Delta p_t] = [(\theta+\phi)^2 + (\theta\rho+\phi)^2 - 2(\theta+\phi)(\theta\rho+\phi)\rho] + \sigma_\varepsilon^2 + 2\sigma_n^2$ and the fraction of the variance attributable to trading frictions is $\pi_1 = \frac{2\sigma_n^2 + [(\theta+\phi)^2 + (\theta\rho+\phi)^2 - 2(\theta+\phi)(\theta\rho+\phi)\rho]}{\sigma_\varepsilon^2 + 2\sigma_n^2 + [(\theta+\phi)^2 + (\theta\rho+\phi)^2 - 2(\theta+\phi)(\theta\rho+\phi)\rho]}$ with the fraction of variance attributable to pricing errors, $\pi_2 = \frac{2\sigma_n^2}{\sigma_\varepsilon^2 + 2\sigma_n^2 + [(\theta+\phi)^2 + (\theta\rho+\phi)^2 - 2(\theta+\phi)(\theta\rho+\phi)\rho]}$.

Estimates of price volatility and its components

	Our Model	Ben Sita and Westerholm	Madhavan et al.
$\text{var}[\Delta p_t]$	0.0247	0.0188	0.0449
π_1	0.9541	0.9427	0.9744
π_2	0.0226	0.0297	0.0012
Liquidity costs	0.0132	0.0149	
Transactions costs	0.0014	0.0012	0.0378
Adverse selection costs	0.0000	0.0010	0.0004

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Table 8: Estimation Results of the Trade model on the Nearby contracts

This table reports the coefficient estimates and corresponding t-statistics of the estimated trade equation:

$$r_t = \left\lfloor \phi_1 \Delta q_t + \left(\phi_2 \psi_t + \rho_q \theta_1 C_t + \phi_3 z_t \right) q_t \right\rfloor + \left(\varepsilon_t + n_t - n_{t-1} \right)$$

Estimation is based on tick-by-tick data for the TIIE28, 28-day interbank interest rate futures contract on MexDer using a sample consisting of 11,687 information ticks on nearby contracts between August 28, 2000 and October 13, 2006.

Trade model – Nearby Contracts		
	Coefficient	**t-statistic**
ϕ_1	0.21699	18.17733
ϕ_2	2.06346	1.345909
ϕ_3	-1.94542	-1.269357
θ_1	3.47964	13.25258

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Table 9 : Estimation Results of Alternative Empirical Models on the nearby contracts

This table reports the coefficient estimates and corresponding t-statistics from the estimation of four alternative empirical models. Panel A reports the Madahavan et al. (1997) model:

$$r_t = \theta\left(q_t - \rho_q q_{t-1}\right) + \phi\left(q_t - q_{t-1}\right) + u_t$$

Panel B reports the Dufour and Engle (2000) model:

$$r_t = \left(\gamma_0 q_t - \gamma_1 q_{t-1}\right) + \left[\delta_0 \ln(x_t) q_t + \delta_1 \ln(x_{t-1}) q_{t-1}\right] + u_t$$

Panel C reports the DeJong et al. (1996) model:

$$r_t = a_0 + \left(R_0 \Delta q_t + R_1 \Delta q_t z_t\right) + \left(e_0 q_{t-1} + e_1 q_t z_{t-1}\right) + u_t$$

Panel D reports the Ben Sita and Westerholm (2006) model:

$$r_t = \left[\phi_1 \Delta q_t + \left(\phi_2 \psi_t + \rho_q \theta_1 C_t\right) q_t\right] + \left(\varepsilon_t + n_t - n_{t-1}\right)$$

Estimation is based on tick-by-tick data for the TIIE28, 28-day interbank interest rate futures contract on MexDer using a sample consisting of 11,687 information ticks on nearby contracts between August 28, 2000 and October 13, 2006.

A: Madahavan et al. (1997) model

	Coefficient	t-statistic
θ	-4.618185	-8.475703
ϕ	4.817981	8.719046

B: Dufour and Engle (2000) model

	Coefficient	t-statistic
γ_0	0.207753	3.527667
γ_1	-0.066664	-1.131929
δ_0	-0.007971	-0.138077
δ_1	-0.010535	-0.182493

C: DeJong et al. (1996) model

	Coefficient	t-statistic
a_0	-0.01492	-1.349278
R_0	0.21555	11.56336
R_1	-0.01502	-0.835671
e_0	0.12784	7.717657
e_1	-0.00790	-0.57123

D: Ben Sita and Westerholm (2006) model

	Coefficient	t-statistic
ϕ_1	0.074439	6.437355
ϕ_2	0.125621	8.674408
θ_1	0.311961	0.433944

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